

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We explore how we can reduce our digital carbon footprint, delve into the mysteries of sleep or rather the lack of it, look into the power source of what drives our spacecraft and more.



Monkeypox – Not another one, please!

As the world was slowly getting rid of the COVID-19 outbreak, a Monkeypox outbreak across several countries has brought to the surface fresh concerns in the minds of the health bodies across the globe. The World Health Organisation is actively monitoring the outbreak and publishing their findings as they come to light. <https://dgit.in/Monkeypox>

WHAT'S NEW

Moon plants – Finally!

Scientists at the University of Florida, conducting a research study funded by NASA, have grown real plants in lunar soil samples. They explored the lunar soil samples brought back to Earth by the Apollo astronauts 50 years ago. The researchers were finally able to grow *Arabidopsis thaliana* in the three of the five samples brought from the Moon.

During the study, the scientists used samples collected by the astronauts during the Apollo 11, 12 and 17 missions, to which they added water, followed by the seeds. To aid the growth of the plants in the nutrient-poor lunar regolith they added a nutrient solution daily. Each plant had 1 gram of lunar regolith for itself. The entire contraption was placed in terrarium boxes in a room to be



Source: NASA

observed and experimented on.

The plants showed first signs of growth after just two days.

By the 16th day, there

were apparent differences in the growth patterns of the plants. Differences were observed in the plants growing in the lunar regolith and the ones growing in the lunar stimulant, which was made using volcanic ash. Further probing into the RNA, which was translated into a protein sequence, suggested that the lunar regolith plants were indeed under stress. While some questions remain unanswered, this finding is a major breakthrough.

<https://dgit.in/MoonPlants>

Gliding salamanders

Salamanders have been known for many traits, but flying wasn't one of them. Until now. A recent study by Christian Brown, a researcher at the University of Florida, has been successful in determining how the wandering salamanders (*Aneides vagrans*) are able to control their flight. It was found that they perform such a feat by using movements similar to those of human skydivers. The findings were published in



Source: USF

a paper co-authored by Christian, titled, *Gliding and parachuting by arboreal salamanders*.

During the process, tests were conducted on four different species of salamanders and the studied traits

included, parachuting with no lateral displacement, gliding, inert falling, etc. Each species of salamander were put in a vertical wind tunnel, where their movements were closely studied and analysed. After about 45 trials, the conclusion was drawn that out of the five species, *A. vagrans* were the ones that demonstrated the best control during their flight. <https://dgit.in/FlyingSalamanders>



Source: CU Boulder

Water from moon volcanoes

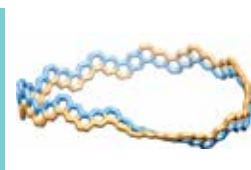
A team of researchers from the University of Colorado at Boulder, have found that below the surface of the Moon, there could be much more water than previously anticipated. It is believed to be in the form of ice sheets formed during volcanic activity in the past. <https://dgit.in/MoonWater>



Source: NASA

High energy pulsars

Scientists have been able to come close to establishing a connection between pulsars, and ultrahigh-energy emissions. This was observed by studying the fast-moving gamma-ray photons that had been discovered earlier using the LHAASO. <https://dgit.in/PulsarEnergy>



Source: Nature

Carbon Möbius strip

Harvesting the complex shape forming ability of carbon, researchers have managed to create a Möbius carbon nanobelt. This, according to the researchers, is a significant step toward enabling other researchers to make much more complex carbon structures. <https://dgit.in/CarbonMobius>